

Analysis First-Year Exam, August 2016, Part 2

Answer exactly *FOUR* questions. Write solutions in a neat and logical fashion, giving complete reasons for all steps.

1. Let the function $f : \mathbb{R} \rightarrow \mathbb{R}$ be uniformly continuous. For each positive integer n let $f_n(t) = f(t + \frac{1}{n})$ whenever t is a real number. Prove that the sequence $(f_n)_{n=1}^\infty$ is uniformly convergent. Give an example (with justification) to show that the conclusion can fail if the hypothesis uniformly is dropped.
2. Let $A = \{a_n : n \geq 1\}$ be a countably infinite subset of $[0, 1]$ and let $1_A : [0, 1] \rightarrow \mathbb{R}$ be its indicator (or characteristic) function. Exhibit such a set A for which 1_A is not Riemann integrable and exhibit such a set A for which 1_A is Riemann integrable, providing justification in each case.
3. Let $(f_n)_{n=1}^\infty$ be a sequence of continuous functions from $[0, 1]$ to $[0, 1]$. Prove that if $f_n(t)$ decreases to 0 whenever $0 \leq t \leq 1$, then $\int_0^1 f_n(t) dt \rightarrow 0$. Does the same conclusion follow when decreases is replaced by converges? Justify.
4. Prove that if the function $f : \mathbb{R} \rightarrow \mathbb{R}$ is differentiable then its derivative is measurable.
5. Let f be a Lebesgue integrable function on $\mathbb{R}$ and let g be defined on $\mathbb{R}$ by $g(y) = \int_{-\infty}^{\infty} \cos(xy)f(x) dx$. Prove that $\lim_{k \rightarrow \infty} g(k) = 0$.